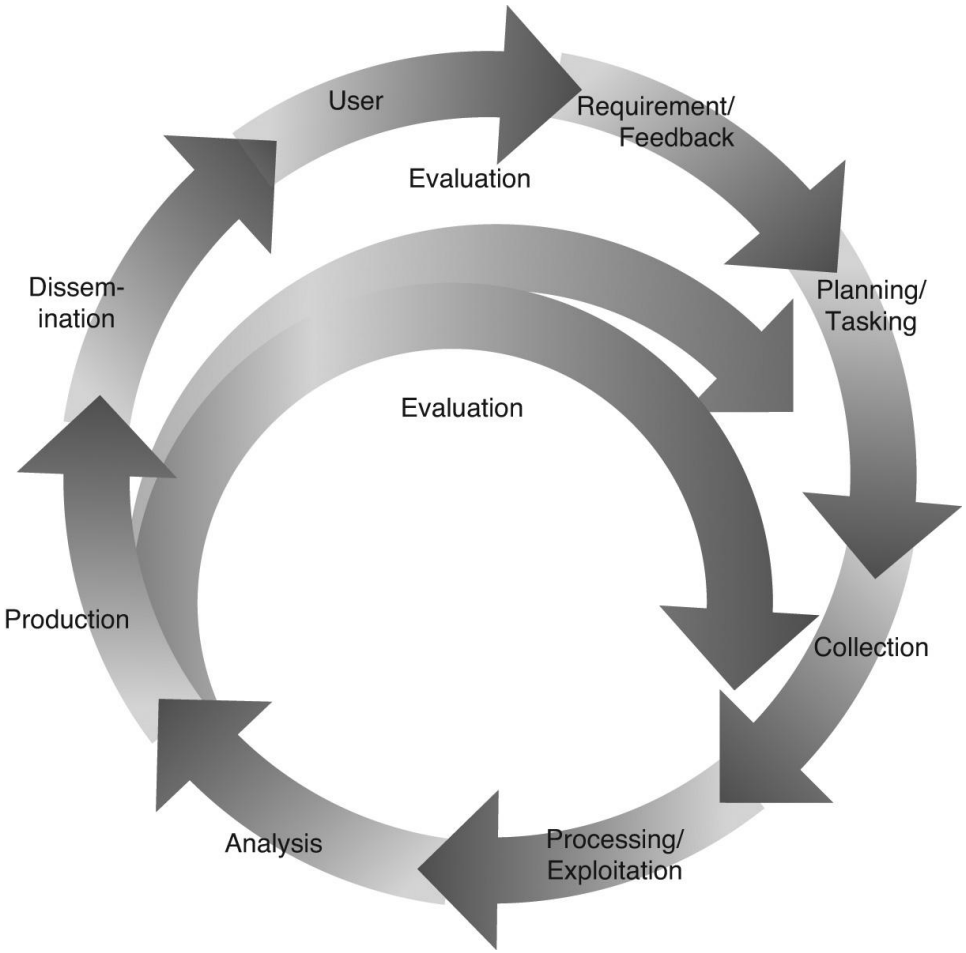
Business Intelligence Initiatives

- Project
  - A collection of temporary business tasks designated to produce a unique product, service or result with a finite duration and resource constraints
- Program
  - A group of related projects managed in a coordinated way to obtain benefits and control not available from managing them individually
- Initiative
  - A collection of projects and programs to achieve specific strategic objectives and goals
    - Goals
      - Long-term outcomes, usually the next three to five years
      - An example – to increase the revenue
    - Objectives
      - Short-term measurable actions to achieve a strategic goal
        - Quantifying and setting the target
      - An example – to increase the revenue by 5% over the next three months

Business Intelligence Initiatives

- BI initiatives are designed to help the organization to achieve its targeted performance
- BI initiatives typically follow a process similar to that used in military intelligence initiatives



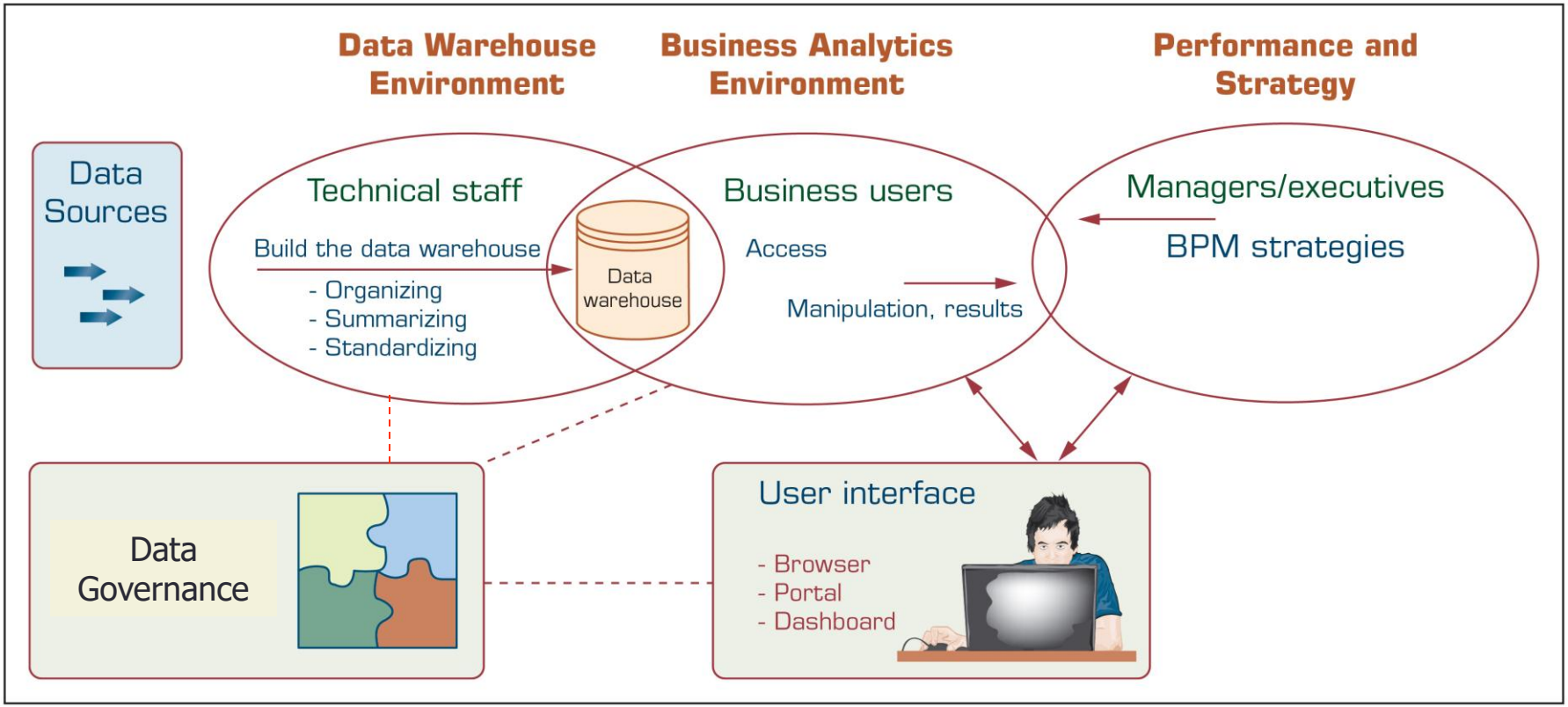
Business Intelligence Initiatives

- 5S
  - Strategy
    - How well does a business intelligence initiative align with the current business/organizational/IT strategies of the organization
  - Scope
    - To what extent does a business intelligence initiative improve particular products, services, projects, programs, etc.
  - Scale
    - How rapidly and cost-effectively can a business intelligence initiative scale up
  - Speed
    - How effective is a business intelligence initiative in accelerating new product launches or the speed of the dynamic supply chain
  - Source of value creation and capture
    - How can a business intelligence initiative create return on investment

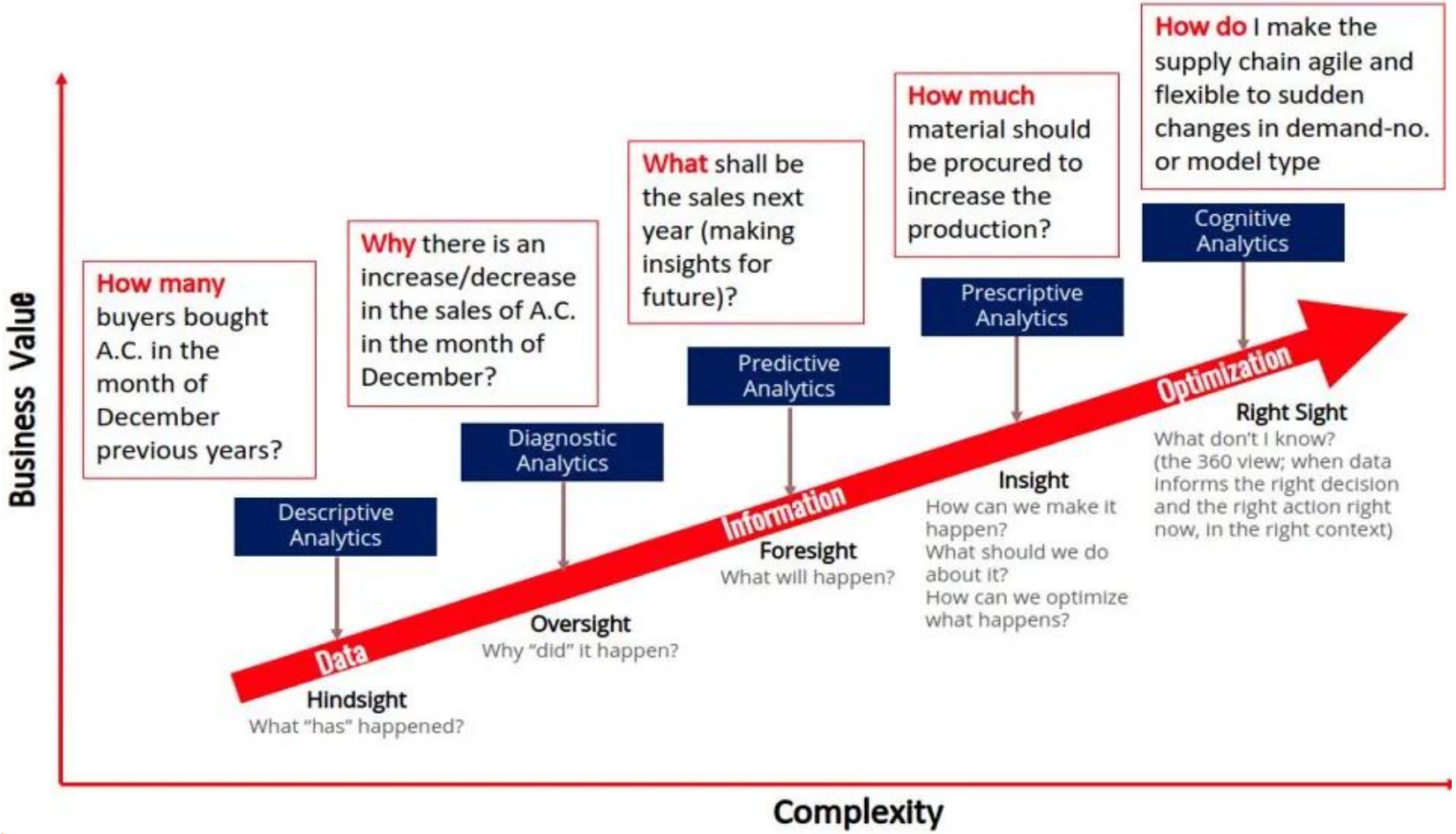
Architecture of BI Systems

- A BI system has the following major components
  - Data warehouse for storing and retrieving data
  - Business analytics for a collection of tools for manipulating and analyzing the data in the data warehouse
  - Business performance management (BPM) for monitoring and analyzing corporate performance
  - User interface for business reporting and data visualization (e.g., dashboards)
  - Data governance for defining the organizational policies, processes and controls that ensure the data are accurate, secure and accessible

Architecture of BI Systems



Business Analytics



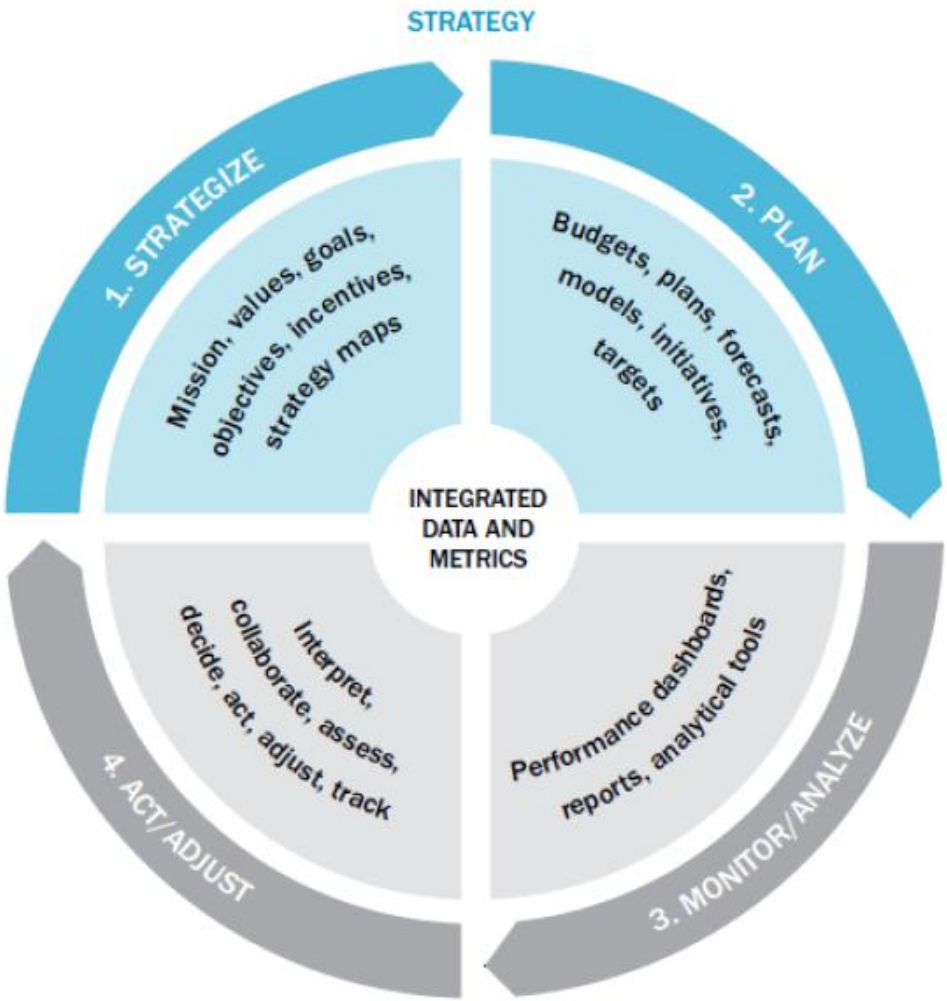
Business Performance Management

- Business Performance Management (BPM)
  - A real-time system that alerts managers to potential opportunities, impending problems and threats, and then empowers them to react through models and collaboration
  - Also called *corporate performance management* (CPM) by Gartner Group, *enterprise performance management* (EPM) by Oracle, *strategic enterprise management* (SEM) by SAP

Business Performance Management

- BPM refers to the business processes, methodologies, metrics, and technologies used by an organization to measure, monitor, and manage its business performance
- BPM encompasses three key components
  - A set of integrated, closed-loop management and analytic processes, supported by technology
  - Tools for businesses to define strategic goals and then measure/manage performance against them
  - Methods and tools for monitoring KPIs, linked to the business, organizational and IT strategies

BPM Loop



- Steps
  1. Strategize
  2. Plan
  3. Monitor/analyze
  4. Act/adjust

Each step with its own sub-steps

Strategize: Where Do We Want to Go?

- Strategic planning
  - Common tasks for the strategic planning process:
    1. Conduct a current situation analysis
    2. Determine the planning horizon
    3. Conduct an environment scan
    4. Identify critical success factors
    5. Complete a gap analysis
    6. Create a strategic vision
    7. Develop a business strategy
    8. Identify strategic objectives and goals

Plan: How Do We Get There?

- Operational planning
  - Operational plan: a plan that translates an organization's strategic objectives and goals into a set of well-defined tactics and initiatives, resource requirements, and expected results for some future time period (usually a year)
- Operational planning can be
  - Tactic-centric (operationally focused)
  - Budget-centric plan (financially focused)

Monitor/Analyze: How Are We Doing?

- A comprehensive framework for monitoring performance should address two key issues:
  - What to monitor
    - Critical success factors
    - Strategic goals and targets
  - How to monitor

Act and Adjust: What Do We Need to Do Differently?

- Success (or mere survival) depends on new projects: creating new products, entering new markets, acquiring new customers (or businesses), or streamlining some process.
- Many new projects and ventures fail!
- What is the chance of failure?
  - 60% of Hollywood movies fail
  - 70% of large IT projects fail

Performance Measurement

- **Performance measurement system**

  A system that assists managers in tracking the implementations of business strategy by comparing actual results against strategic goals and objectives

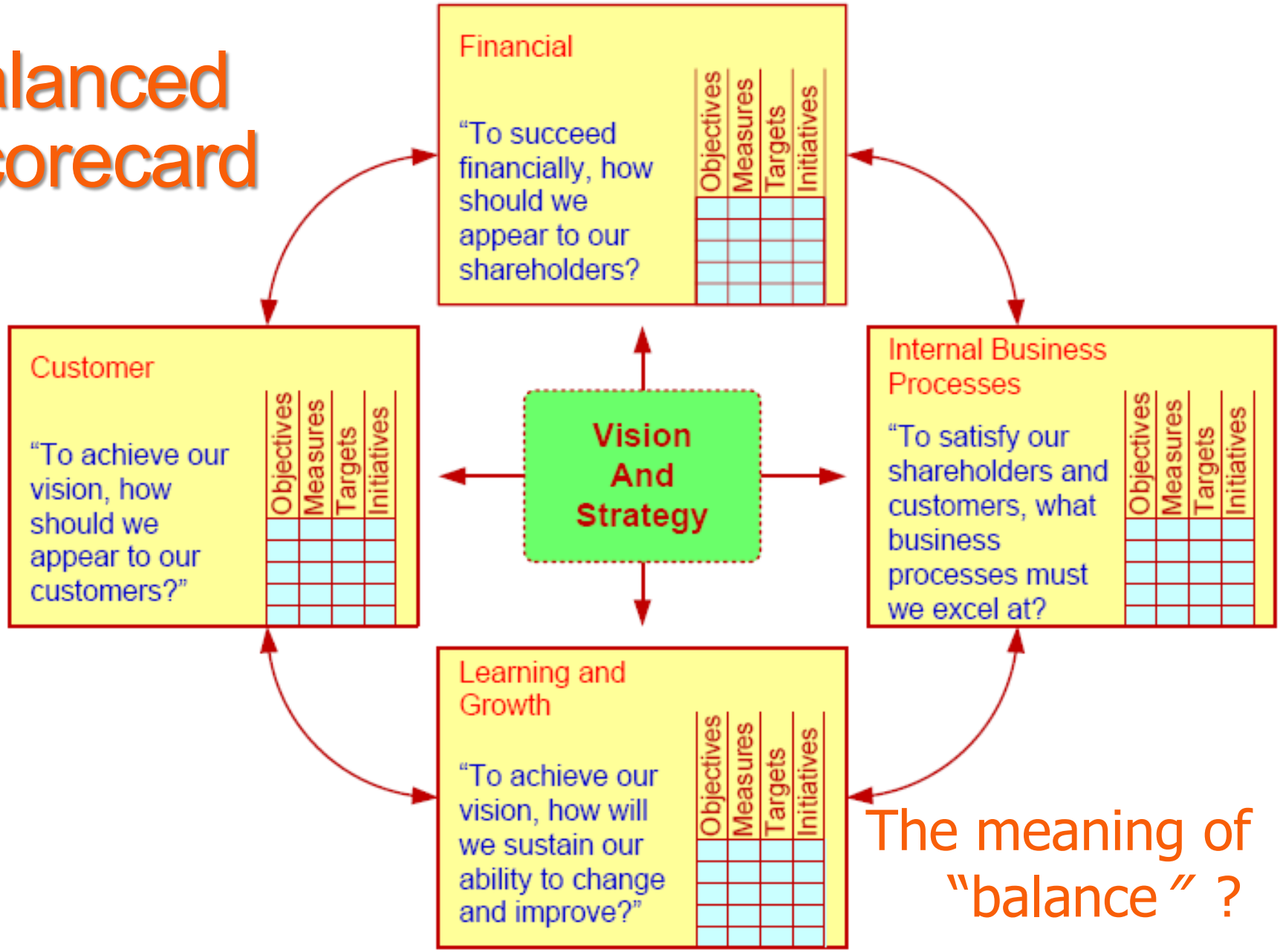
  - Comprises systematic comparative methods that indicate progress (or lack thereof) against goals

Performance Measurement

- **Balanced Scorecard**

  A performance measurement and management methodology that helps translate an organization's financial, customer, internal process, and learning and growth objectives and targets into a set of actionable initiatives

Balanced Scorecard



The meaning of "balance" ?

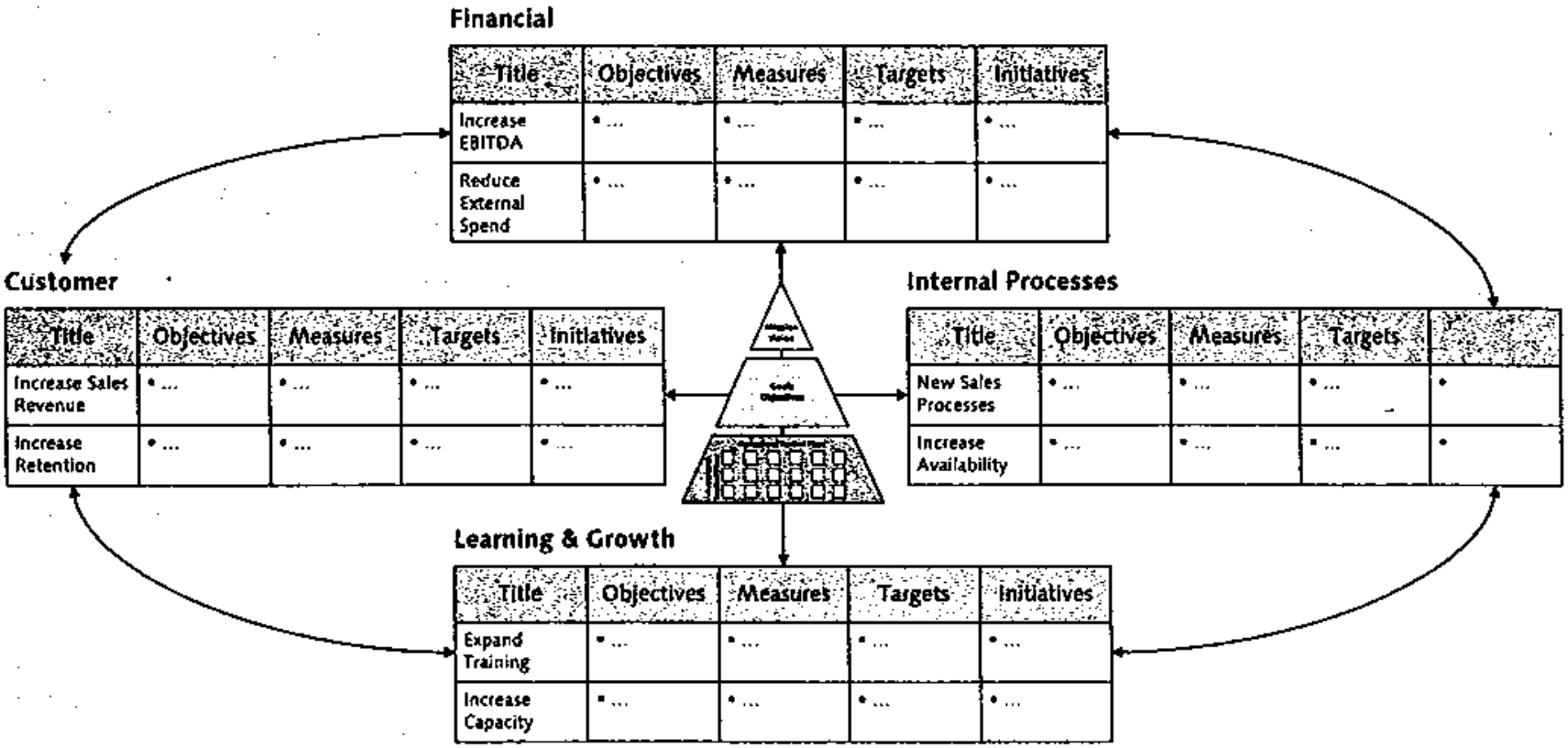
Balanced Scorecard

- Performance measurement levels
  - At the corporate level (financial perspective)
  - At the division (profit center) level (customer perspective)
  - At the department (cost center) level (internal business process perspective)
  - At the individual level (learning & growth perspective)

Balanced Scorecard

Financial Goals 1) Improve ROI 2) Increase Revenue (Turnover) 3) Optimize Cost Efficiency 4) Optimize Capital employed 5) Increase net liquidity	**Customer Goals** 6) Building long-term Partnerships 7) Firm Branding 8) Become full service provider for each of our customers 9) Become China's leading IT provider for automotive and discrete manufacturing industry 10) Double market share automotive and discrete manufacturing industry 11) Increase share of client ABC IT-Budget 12) Customer Satisfaction
Process Goals 13) Implement common Portfolio and Sales Process 14) Implement common order management, delivery process and resource management 15) Implement common Finance and Controlling process 16) Plan, build, run the gedas process model 17) Innovation management and Portfolio Management	**Potential Goals** 18) Attractive company for professionals 19) Skill Management 20) Describe the firm's values and culture 21) Communicate corporate objectives

Balanced Scorecard



Financial Perspective

- How does the firm look to shareholders?
  - Revenue growth and mix
  - Cost reduction/productivity improvement
  - Asset utilization/investment strategy
- Improved business performance pays off only when it results in
  - Improved sales
  - Reduced operating costs
  - Faster asset turnover
  - Higher stock price
- If financial results are poor, review the business, IT and organizational strategies

Financial Perspective

- Increase economic value through revenue growth and productivity
  - Revenue growth
    - Sales to new markets, customers or with new products
    - More sales to existing customers by deepening relationships (cross-selling and up-selling)
  - Productivity
    - Improve cost structure by lowering direct and indirect expenses
    - Utilize assets more efficiently by reducing working and fixed capital

Customer Perspective

- How can the firm create value for customers?
  - Time from product order to delivery
  - Time to get the product from the definition stage to market
  - On-time delivery measures
  - Defect level as measured by customers
- The core issue is the customer value proposition
  - Unique mix of price, product and service
  - Customer experience and relationship (loyalty)
  - Reputation and image that the firm offers

Customer Perspective

- Expected outcomes from the customer value proposition
  - Market share in targeted customer segments
  - Acquisition and retention of customers in targeted segments
  - Targeted customer segments' profitability
- Examples of the customer value proposition
  - Operational excellence in terms of the lowest price and/or exceptional service (e.g., Walmart, McDonald's)
  - Product leadership in terms of functionalities, features and performance (e.g., Sony, Dyson)

Internal Business Process Perspective

- What must the firm do in order to meet the customers' needs?
  - Productivity
  - Efficiency
  - Quality
- How will the firm achieve its differentiated value proposition?
  - Build the franchise
  - Increase customer value
  - Achieve operational excellence
- Not just cost/time/quality

Learning and Growth Perspective

- Continue improving employee capabilities and skills
- Continue creating value when the targets for success keep changing
  - Identify the gaps between the existing capabilities of people and those needed to achieve the targets
    - Employee satisfaction, retention and skills
    - Innovations: new products or services

Performance Measurement

- Six Sigma
  - A performance management methodology aimed at reducing the number of defects in a business process to as close to zero (<3.4) defects per million opportunities (DPMO) as possible
    - The most popular Total Quality Management (TQM) technique
    - Originally developed at Motorola
    - Institutionalized at GE for “near-perfect products”

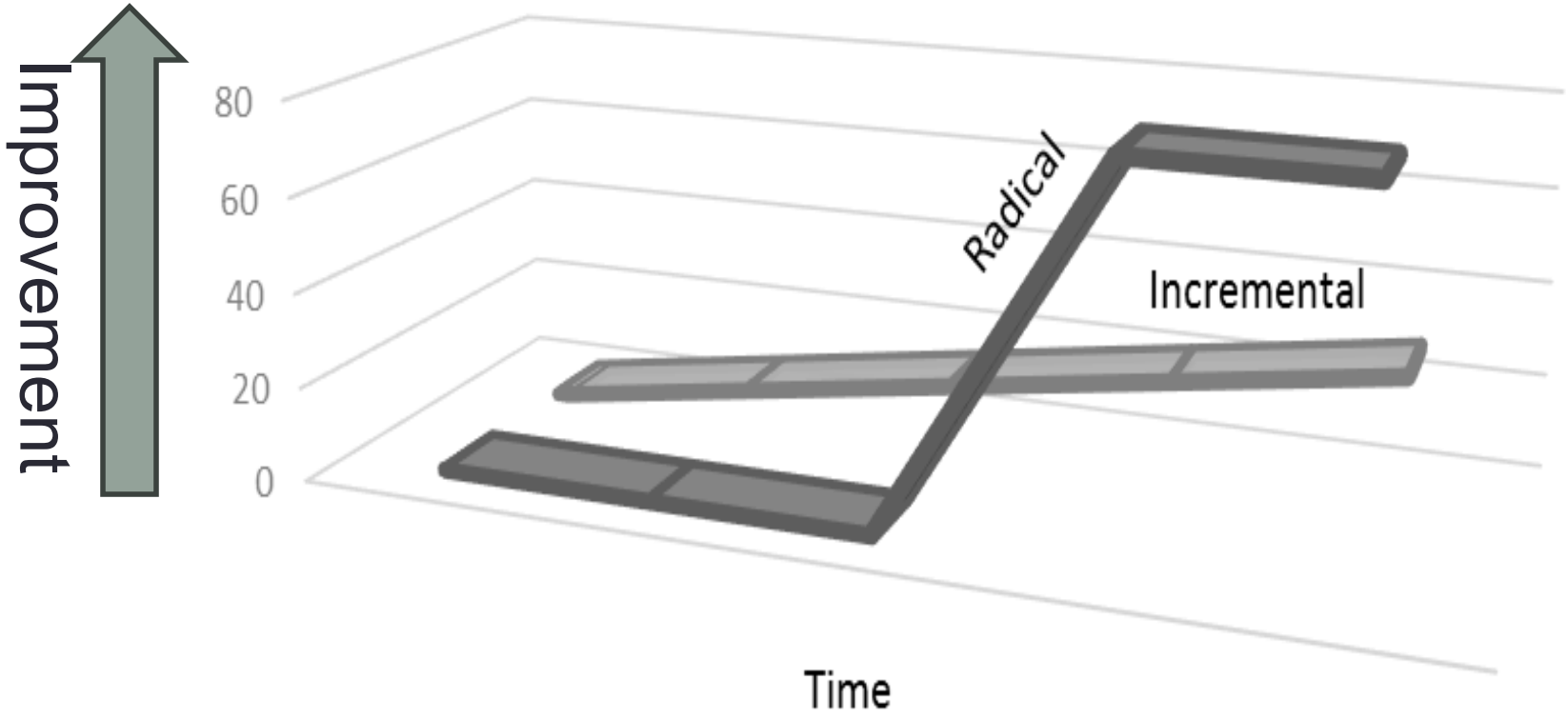
Performance Measurement

- Six Sigma follows the DMAIC performance model
  - A closed-loop business improvement model that encompasses the steps of defining, measuring, analyzing, improving, and controlling business processes

Changing Business Processes

- Total Quality Management (TQM)
  - Incremental, continuous process improvement
  - Often results in favorable reactions from personnel
    - Improvements are owned and controlled
    - Less threatening change
  - Six Sigma is the most popular TQM technique
- Business Process Reengineering (BPR)
  - Sets aggressive improvement goals
  - Goal is to make a rapid, breakthrough impact on KPIs in a short amount of time
  - Greater resistance by personnel
    - Use only when radical change is needed
  - ERP is the most popular BPR technique

Comparison of TQM and BPR



SMART KPI Design

- KPIs can measure the success of anything
  - They can apply to projects, programs, initiatives, products, services, etc.
  - They should be tailored to specific objectives
- SMART is a goal-setting framework for designing KPIs
  - **S**pecific, **M**easurable, **A**chievable, **R**elevant, and **T**ime-bound

SMART KPI Design

- An example of the KPI for sales leads
  - What: Increase the sales leads by 30% this year
  - Why: Achieving this target will increase the profit
  - How: Hiring additional sales staff, improving existing marketing strategies, adopting a new CRM system, etc.
  - Who: VP of Sales is responsible for this KPI
  - When: The KPI will be reviewed and analyzed on a quarterly basis

OKR

- Objectives and Key Results (OKR) is another goal-setting framework for defining and tracking objectives and their outcomes
- The purpose of OKRs is to define how to achieve objectives through concrete, specific and measurable actions
- OKRs are typically set at the organization, team and individual levels

OKR

- OKR = 1 objective and 3-5 key results
  - Objective = a clearly defined target
    - The objective should be supported by initiatives, which are the plans and activities that help to achieve the objective and move forward the key results
  - Key results = specific measurements used to track the achievement of the objective
    - They can be measured on a 0-100% scale or any numerical unit (e.g., dollar amount, the number of items, etc.)

OKR

- An example of the OKR at the organizational level
  - Objective: Increase brand recognition and awareness
  - Key result 1: Increase media engagement by 20%
  - Key result 2: Launch the customer referral program by December 1
  - Key result 3: Extend social media reach and visibility to 2 new target markets

OKR

- An example of the OKR at the team level
  - Objective: Increase social media engagement by 30%
  - Key result 1: Identify the 3 most popular social media sites among the 2 new target audiences and develop the engagement strategy by December 1
  - Key result 2: Respond to new Facebook comments within 3 hours
  - Key result 3: Increase the number of followers on Facebook by 20%

OKR

- An example of the OKR at the individual level
  - Objective: Increase the number of social media connections by 30%
  - Key result 1: Triple the posting frequency on Facebook
  - Key result 2: Establish presence on the 2 new social media sites: Linkedin and Quora
  - Key result 3: Join 5 Linkedin groups and leave comments on the 10 most popular discussions in each group
  - Key result 4: Gain 10 followers on Quora by posting 3 questions/answers every week

OKR

- Comparing with KPIs, OKRs are better for focusing on larger visions and creating sustainable changes in terms of business performance
- OKRs are often aggressive and ambitious
  - Aggressive OKRs can push the team or individual to perform better
  - They should not be unreachable

OKR

The Road Trip

Where you want to go?

Travel Guide

Strategy

Are you on the right path?

GPS

OKR

Is everything else OK?

Dashboard

Monitoring KPIs